

Learning outcomes

By the end of this topic, you will be able to:

7.1 Progressive waves

- 1 describe what is meant by wave motion as illustrated by vibration in ropes, springs and ripple tanks
- 2 understand and use the terms displacement, amplitude, phase difference, period, frequency, wavelength and speed
- 3 understand the use of the time-base and y-gain of a cathode-ray oscilloscope (CRO) to determine frequency and amplitude
- 4 derive, using the definitions of speed, frequency and wavelength the wave equation $v = f\lambda$
- 5 recall and use the equation $v = f\lambda$
- 6 understand that energy is transferred by a progressive wave
- 7 recall and use intensity = power/area and intensity is proportional to (amplitude)² for a progressive wave

7.2 Transverse and longitudinal waves

- 1 compare transverse and longitudinal waves
- 2 analyse and interpret graphical representations of transverse and longitudinal waves

7.3 Doppler effect for sound waves

- 1 understand that when a source of sound waves moves relative to a stationary observer, the observed frequency is different from the source frequency
- 2 use the expression $f_o = f_s v / (v \pm v_s)$ for the observed frequency when a source of sound waves moves relative to a stationary observer

7.4 Electromagnetic spectrum

- 1 state that all electromagnetic waves are transverse waves that travel with the same speed c in free space
- 2 recall the approximate range of wavelengths in free space of the principal regions of the electromagnetic spectrum from radio waves to γ -rays
- 3 recall that wavelengths in the range 400–700 nm in free space are visible to the human eye

7.5 Polarisation

- 1 understand that polarisation is a phenomenon associated with transverse waves
- 2 recall and use Malus's law ($I = I_0 \cos^2 \theta$) to calculate the intensity of a plane polarised electromagnetic wave after transmission through a polarising filter or a series of polarising filters

Starting points

- ★ The basic properties of waves, such as reflection, refraction, diffraction and interference.
- ★ The electromagnetic spectrum.
- ★ Visible light is a small part of the electromagnetic spectrum with each colour having a different frequency.

**7.1 Progressive waves: transverse and longitudinal waves**

This topic will introduce some general properties of waves. We will meet two broad classifications of waves, **transverse** and **longitudinal**, based on the direction in which the particles vibrate relative to the direction in which the wave transmits energy. We will

define terms such as amplitude, wavelength and frequency for a wave, and derive the relationship between speed, frequency and wavelength. We will look at demonstrations of some properties of waves, such as reflection and refraction.

Wave motion is a means of moving energy from place to place. For example, **electromagnetic waves** from the Sun carry the energy that plants need to survive and grow. The energy carried by sound waves causes our ear drums to vibrate. The energy carried by seismic waves (earthquakes) can devastate vast areas, causing land to move and buildings to collapse. Waves which transfer energy from place to place without the transfer of matter are called **progressive waves**.

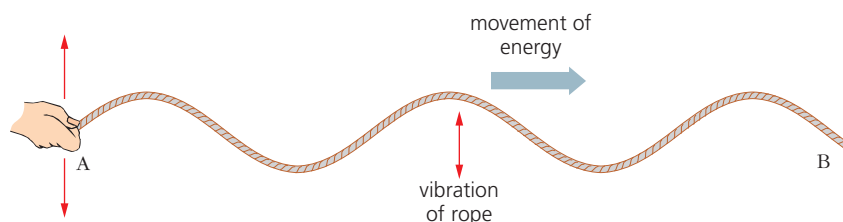
Vibrating objects act as sources of waves. For example, a vibrating tuning fork sets the air close to it into oscillation, and a sound wave spreads out from the fork. For a radio wave, the vibrating objects are electrons.

There are two main groups of waves. These are **transverse** waves and **longitudinal** waves.

A transverse wave is one in which the vibrations of the particles in the wave are at right angles to the direction in which the energy of the wave is travelling.



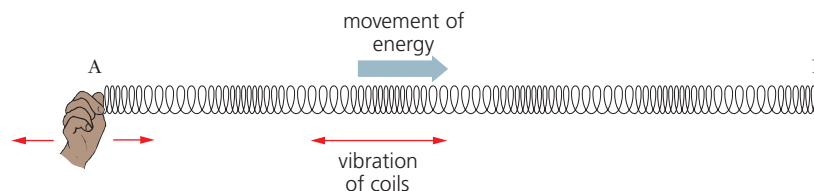
Figure 7.1 shows a transverse wave moving along a rope. The particles of the rope vibrate up and down, while the energy travels at right angles to this, from A to B. There is no transfer of matter from A to B. Examples of transverse waves are electromagnetic waves which include light waves, surface water waves and secondary seismic waves (S-waves).



▲ **Figure 7.1** Transverse wave on a rope

A longitudinal wave is one in which the direction of the vibrations of the particles in the wave is along or parallel to the direction in which the energy of the wave is travelling.

Figure 7.2 shows a longitudinal wave moving along a stretched spring (a 'slinky'). The coils of the spring vibrate along the length of the spring, whilst the energy travels along the same line, from A to B. Note that the spring itself does not move from A to B. Examples of longitudinal waves include sound waves and primary seismic waves (P-waves).



▲ **Figure 7.2** Longitudinal wave on a slinky spring

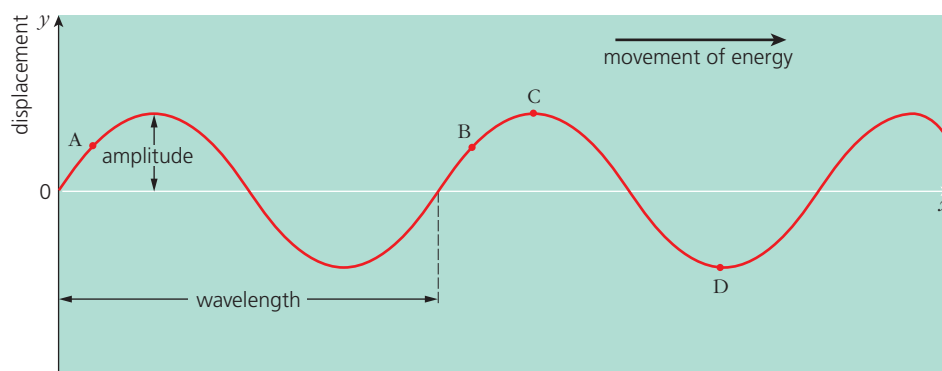


Graphical representation of waves

The **displacement** of a particle in a wave is its distance in a specified direction from its rest/equilibrium position.

Displacement is a vector quantity; it can be positive or negative. A transverse wave may be represented by plotting displacement y on the y -axis against distance x along the wave, in the direction of energy travel, on the x -axis. This is shown in

Figure 7.3. It can be seen that the graph is a snapshot of what is actually observed to be a transverse wave. The crest at C and the trough at D will be seen to move to the right with time transferring energy to the right. The particles at C and D will move towards zero displacement at right angles to the direction the energy is travelling.



▲ **Figure 7.3** Displacement–distance graph for a transverse wave

For a longitudinal wave, the displacement of the particles is along or parallel to the direction of energy travel. However, if these displacements are plotted on the y -axis of a graph of displacement against distance, the graph has exactly the same shape as for a transverse wave (see Figure 7.3). This is very useful, in that one graph can represent both types of wave. Using this graph, wave properties may be treated without reference to the type of wave.

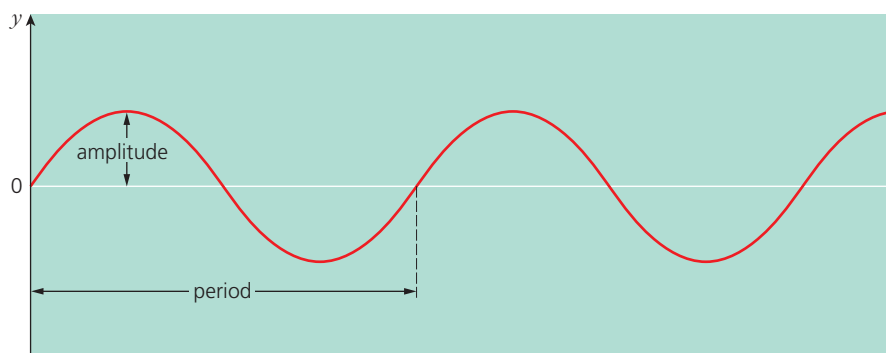
The **amplitude** of the wave motion is defined as the maximum displacement of a particle in the wave from its rest/equilibrium position.

It can be seen in Figure 7.3 that the wave repeats itself after a certain distance. That is, the wave can be constructed by repeating a particular section of the wave. The smallest distance that shows the section of the wave that is repeated is called the **wavelength**.

The motion of any particle in the wave from the maximum positive displacement (a crest) to a maximum negative displacement (a trough) back to a maximum positive displacement is called a **cycle** or **oscillation**. Note that the distance moved by a particle in the wave from crest to trough is twice the amplitude and in one oscillation the particle moves a distance that is four times the amplitude.

Another way to represent both waves is to plot a graph of displacement y of a particle in the wave against time t . This is shown in Figure 7.4. Again, the wave repeats itself, in this case after a certain interval of time. This is the time for one complete cycle or oscillation and is called the **period** T of the wave.

The **period** of the wave is the time for a particle in the wave to complete one oscillation or one cycle.



▲ **Figure 7.4** Displacement–time graph for a wave

The number of oscillations (cycles) per unit time is called the **frequency** f of the wave.

The displacement and amplitude are measured in mm, m or other units of length. Period is measured in seconds (s). Frequency has the unit per second (s^{-1}) or **hertz (Hz)**. The period $T = 1/f$.

A term used to compare the displacements and relative motions of particles in a wave is **phase difference**. Two particles that vibrate together are said to be **in phase**. In Figure 7.3 the particles at A and B move in phase as they have the same displacements at the same times and are moving in the same direction. Both particles will move down towards the rest position (zero displacement) as the wave moves towards the right. Particles C and D are half a cycle out of phase as C is at a crest when D is at a trough. Particle C moves down towards the rest position (zero displacement) and particle D moves up towards the rest position. The phase difference is generally stated as an angle expressed in radians or degrees.

One cycle in the wave is represented by 2π radians or 360° .

The radian is included in the following sections as angles and phase differences may be given in this unit. In AS Level, questions are set where angles and phase differences are given in degrees, and answers may be given in degrees or radians. The radian is defined in Topic 12.1.

To convert from a fraction of a cycle to an angle the fraction is multiplied by 2π for radians or 360° for degrees. Hence in Figure 7.3 C and D have a phase difference of π radians or 180° .

The **wavelength** is the minimum distance between particles which are vibrating in phase with each other.

The wavelength is, therefore, the minimum distance between two crests or two troughs.

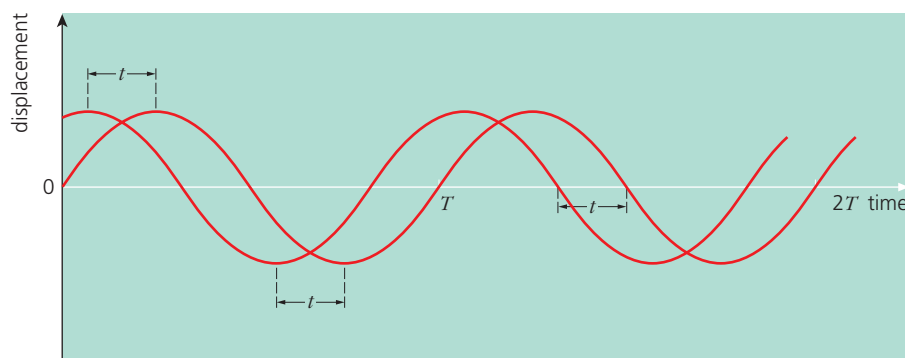
For two-dimensional waves the term **wavefront** is used to join points that are in phase. The ripples seen on the surface of water in a ripple tank are wavefronts (see Figures 7.6 and 7.7 on page 116).

The wavelength is also the distance between adjacent wavefronts. It is the distance moved by the wavefront or the energy during one cycle/oscillation of the source of the waves.

The usual symbol for wavelength is λ , the Greek letter lambda.

The term phase difference is also used to describe the relative positions of the crests or troughs of two *different* waves of the same frequency. When the crests and troughs of the two waves are aligned, the waves are said to be in phase. When crests and troughs are not aligned the waves are said to have a phase difference. When a crest and a trough of two waves are aligned the waves are said to be in **antiphase**. Thus, when waves are out of phase with a crest aligned with a trough, one wave is half a cycle behind the other. In this case, the phase difference between waves that are exactly out of phase (in antiphase) is π radians or 180° . (Phase difference is also discussed in Topic 8 for the superposition of two progressive waves.)

Consider Figure 7.5 (overleaf), in which there are two waves of the same frequency, but with a phase difference between them. The period T corresponds to a phase angle of 2π rad or 360° . The two waves are out of step by a time t . Thus, phase difference is equal to $2\pi(t/T)$ rad = $360(t/T)^\circ$. A similar argument may be used for waves of wavelength λ which are out of step by a distance x . In this case the phase difference is $2\pi(x/\lambda)$ rad = $360(x/\lambda)^\circ$.



▲ Figure 7.5 Phase difference

WORKED EXAMPLE 7A

In Figure 7.5 the time period T is 2.0 s and the time t is 0.25 s. Calculate the phase difference between the two waves in degrees.

Answer

The fraction of time t to the period T is $0.25/2 = 0.125$.

Hence the phase difference is $0.125 \times 360 = 45^\circ$.

Question

- Two waves of the same frequency have a time period of 3.0 s. The two waves are out of phase by a time difference of 0.5 s. Calculate the phase difference between the two waves in degrees.



Wave equation

The speed of a progressive wave is the distance moved per unit time by the wavefronts (or crests) or wave energy. The definition of wavelength λ is the minimum distance between wavefronts. Hence in one cycle of the source the wave energy moves a distance λ . The time taken for one cycle is the time period T . Referring to Topic 2, speed v is the distance moved per unit time. Therefore,

$$v = \lambda/T$$

If f is the frequency of the wave, then $f = 1/T$. Therefore,

$$v = f\lambda$$

or

$$\text{speed} = \text{frequency} \times \text{wavelength}$$

This is an important relationship between the speed of a wave and its frequency and wavelength.

Intensity

One of the characteristics of a progressive wave is that it carries energy. The amount of energy passing through unit area per unit time is called the **intensity** of the wave.

The intensity of a wave is the power per unit area.

The intensity is proportional to the square of the amplitude of a wave for constant frequency. Thus, doubling the amplitude of a wave increases the intensity of the wave by a factor of four. The intensity also depends on the frequency: intensity is proportional to the square of the frequency.

For a wave of frequency f and amplitude A , the intensity I is proportional to $A^2 f^2$.

If the waves from a point source spread out equally in all directions, a spherical wave is produced. As the wave travels further from the source, the energy it carries passes through an increasingly large area. Since the surface area of a sphere is $4\pi r^2$, the intensity is $W/4\pi r^2$, where W is the power of the source. The intensity of the wave thus decreases with increasing distance from the source. The intensity I is proportional to $1/r^2$, where r is the distance from the source.

This relationship assumes that there is no absorption of wave energy.

WORKED EXAMPLE 7B

- 1 A tuning fork of frequency 170 Hz produces sound waves of wavelength 2.0 m. Calculate the speed of sound.
- 2 The amplitude of a wave in a rope is 15 mm. If the amplitude were changed to 20 mm, keeping the frequency the same, by what factor would the power carried by the rope change?

Answers

- 1 Using $v = f\lambda$, we have $v = 170 \times 2.0 = 340 \text{ m s}^{-1}$.
- 2 Intensity is proportional to the square of the amplitude. Here the amplitude has been increased by a factor of 20/15, so the power carried by the wave increases by a factor of $(20/15)^2 = 1.8$.

Questions

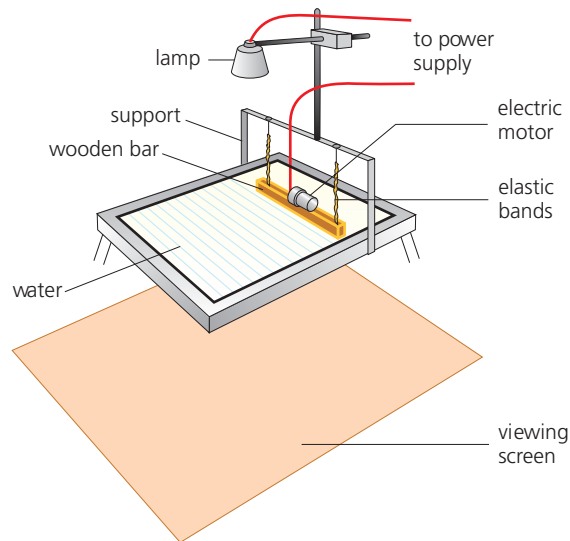
- 2 Water waves of wavelength 0.080 m have a frequency 5.0 Hz. Calculate the speed of these water waves.
- 3 The speed of sound is 340 m s^{-1} . Calculate the wavelength of the sound wave produced by a violin when a note of frequency 500 Hz is played.
- 4 A sound wave has twice the intensity of another sound wave of the same frequency. Calculate the ratio of the amplitudes of the waves.

Properties of wave motions

Although there are many different types of waves (light waves, sound waves, electromagnetic waves, mechanical waves, etc.) there are some basic properties which they all have in common. All waves can be reflected and refracted. All waves can be diffracted, and can produce interference patterns (diffraction and interference will be discussed in Topic 8).

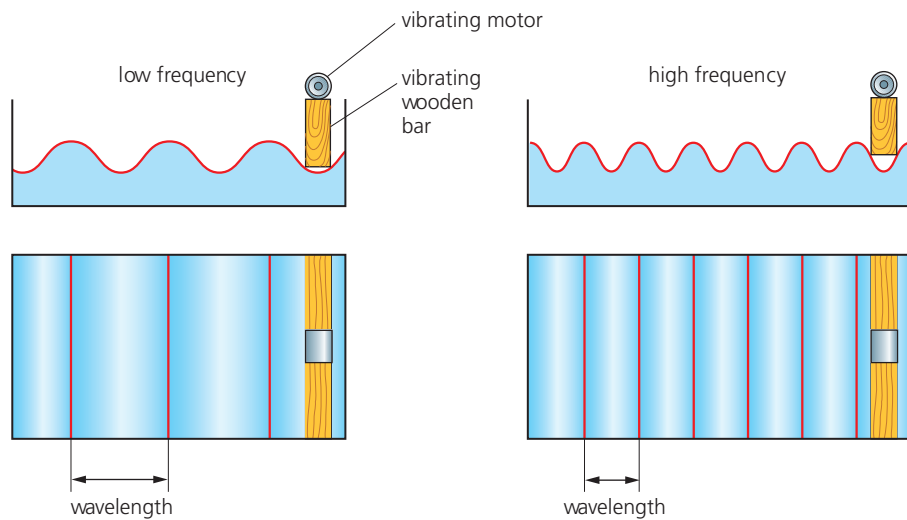


These properties may be demonstrated using a ripple tank similar to that shown in Figure 7.6 (overleaf). As the motor turns, the wooden bar vibrates, creating ripples on the surface of the water. A lamp shines light onto the ripples from above. This creates shadows of the waves on the viewing screen below the tank. The shadows show the shape and movement of the waves. Each dark shadow line joins points that are in phase. These lines represent a **wavefront**. The minimum distance between the wavefronts is the wavelength of the water waves. A stroboscope can be used to 'freeze' the pattern on the screen. The stroboscope emits flashes of light at a given frequency. The frequency is adjusted until it is the same as the frequency of the waves and the shadows are seen in fixed positions.



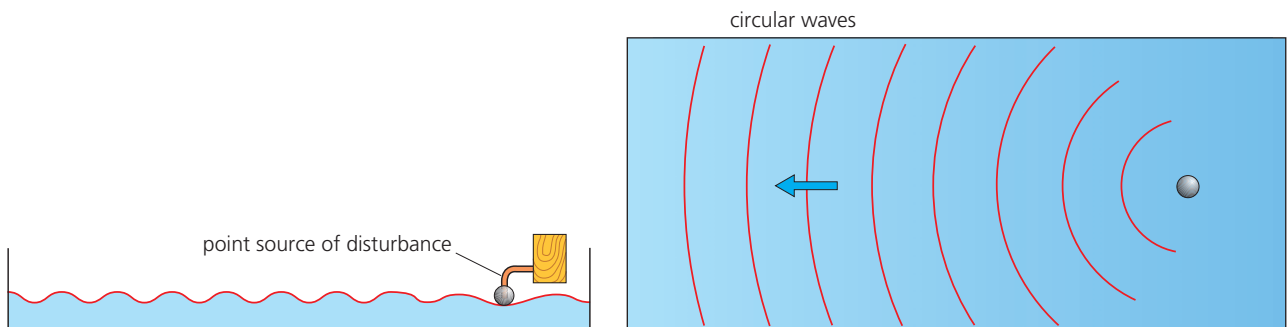
▲ **Figure 7.6** Ripple tank

Figure 7.7 illustrates the pattern of wavefronts produced by a low-frequency vibrator and one of higher frequency. Note that for the higher frequency the wavelength is shorter, since wave speed is constant and $v = f\lambda$.



▲ **Figure 7.7** Ripple tank patterns for low and high frequency vibrations

Circular waves may be produced by replacing the vibrating bar with a small dipper, or by allowing drops of water to fall into the ripple tank. A circular wave is illustrated in Figure 7.8. This pattern is characteristic of waves spreading out from a point source.

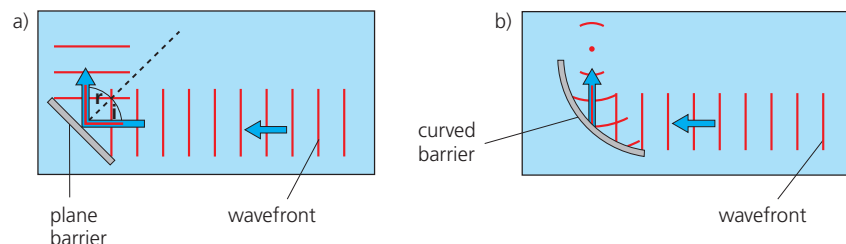


▲ **Figure 7.8** Ripple tank pattern for a point source

We shall now see how the ripple tank may be used to demonstrate the wave properties of reflection and refraction.

Reflection

As the waves strike a plane barrier placed in the water, they are reflected. The angle of reflection equals the angle of incidence, and there is no change in wavelength (see Figure 7.9a). If a curved barrier is used, the waves can be made to converge or diverge (Figure 7.9b).

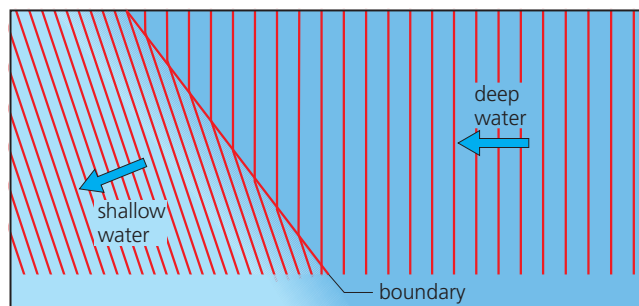


▲ Figure 7.9 Ripple tank pattern showing reflection at a) a plane surface and b) a curved one

Refraction

If a glass block is submerged in the water, this produces a sudden change in the depth of the water. The speed of surface ripples on water depends on the depth of the water: the shallower the water, the slower the speed. Thus, the waves move more slowly as they pass over the glass block. The frequency of the waves remains constant, and so the wavelength decreases. If the waves are incident at an angle to the submerged block, they will change direction, as shown in Figure 7.10.

The change in direction of a wave due to a change in speed is called **refraction**.

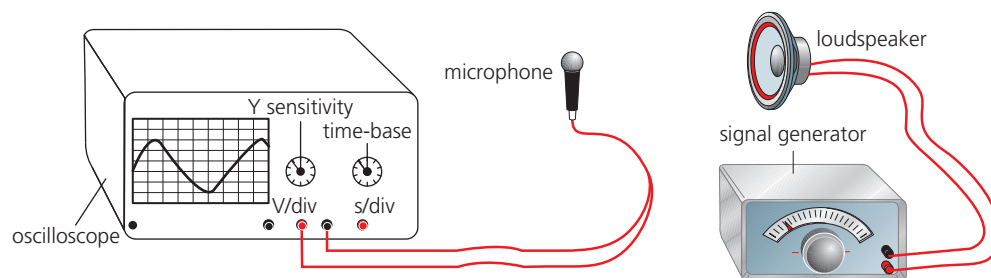


▲ Figure 7.10 Ripple tank pattern showing refraction



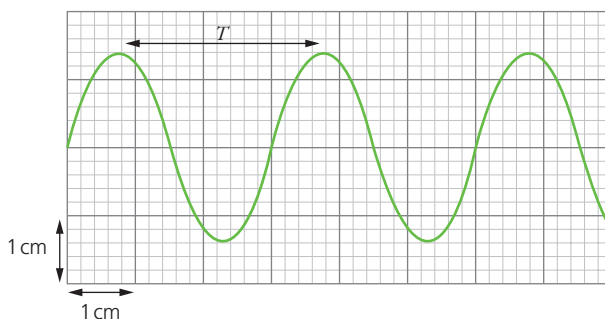
The determination of the frequency and amplitude of sound using a calibrated CRO

A cathode ray oscilloscope (CRO) with a calibrated time-base may be used to determine the frequency of sound. A method of measuring the frequency of sound waves is illustrated in Figure 7.11. A signal generator and loudspeaker are used to produce a note of a single frequency. The microphone is connected to the Y-plates of the CRO.



▲ Figure 7.11 Measuring the frequency and amplitude of sound using a CRO

The microphone detects the sound and a trace on the CRO can be obtained by adjusting the Y-plate sensitivity and the time-base settings. A typical trace is shown in Figure 7.12.



▲ **Figure 7.12** Measurement of the time period and amplitude from the CRO trace

The distance between peaks or troughs is measured using the scale on the x-axis on the CRO display. The time-base setting is used to determine the time period T and frequency of the sound. The calculated value can be compared with that shown on the signal generator.

The amplitude of the sound is represented by the amplitude of the trace on the CRO. The distance from crest to trough along the y-axis is measured and divided by two. The Y-sensitivity, usually given in V cm^{-1} , is used to determine the amplitude of the wave shown on the CRO in volts. The intensity of different sounds can be compared by the value obtained for the amplitude on the CRO. Provided the detector produces a voltage that is proportional to the intensity of the sound, the amplitude of the trace on the CRO is directly related to the intensity. For constant frequency the intensity is proportional to the (amplitude)². For example, if the intensity of the sound wave is doubled, the amplitude of the sound wave is increased by the square root of two and the trace on the CRO shows the same increase.

The CRO can be used to measure the frequency and compare amplitudes of other waves using a suitable detector in place of the microphone.

WORKED EXAMPLE 7C

- The time-base setting for the CRO used to obtain the trace in Figure 7.12 is 2.0 ms cm^{-1} . Determine for the sound:
 - the time period
 - the frequency.
- The Y-sensitivity for the CRO is set on 2.0 V cm^{-1} . Determine the voltage amplitude of the trace shown on the CRO.

Answers

- The distance for a time period is 3.0 cm.
Hence the period = $3.0 \times 2.0 \times 10^{-3} = 6.0 \times 10^{-3} \text{ s}$
 - The frequency = $1/\text{time period} = 1/6.0 \times 10^{-3} = 170 \text{ Hz}$
- The amplitude of the trace is 1.4 cm and hence the voltage amplitude is $1.4 \times 2.0 = 2.8 \text{ V}$.

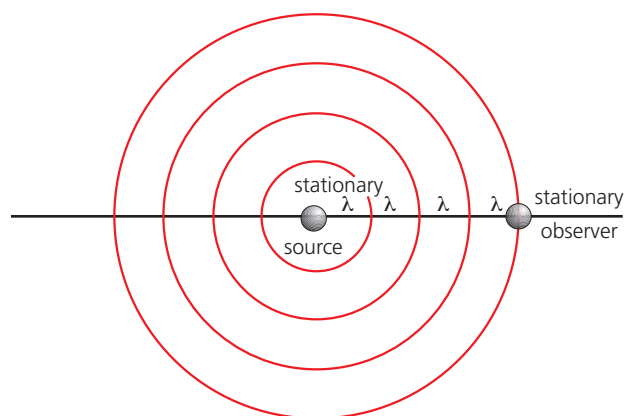
- 5 The time-base on a CRO is set on 0.50 ms cm^{-1} . The trace obtained for a sound wave shows three complete time periods in 7.2 cm. Calculate:
 - a the time period
 - b the frequency.
- 6 Make a copy of the trace shown in Figure 7.12. Draw a second trace on your copy for a sound wave that has half the intensity of the original sound but with the same frequency.



7.2 Doppler effect

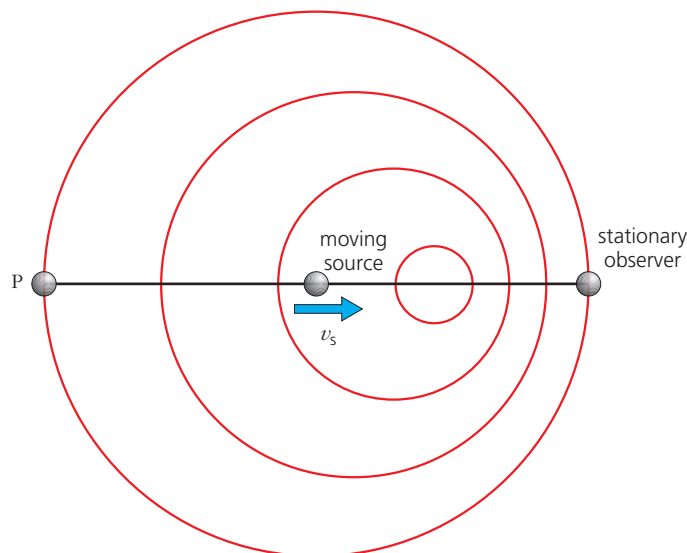
The whistle of a train or the siren of a police car appears to increase in frequency (pitch) as it moves towards a stationary observer. The frequency change due to the relative motion between a source of sound or light and an observer is known as the **Doppler effect**.

When the observer and source of sound are both stationary, the number of waves per second reaching the observer will be the same frequency as the source (see Figure 7.13).



▲ **Figure 7.13** The source emits waves of wavelength λ . The observer is stationary and receives waves with the same wavelength λ .

When the source moves towards the observer the effect is to shorten the wavelength of the waves reaching the observer (see Figure 7.14).



▲ **Figure 7.14** Source of sound moving towards a stationary observer

Let v be the speed of sound in air. A source of sound has a frequency f_s and wavelength λ . The source moves *towards* an observer at a speed v_s .

The period of oscillation of the source of sound is $T (= 1/f_s)$. In the time of one oscillation the source moves **towards** the observer a distance $v_s T$. Hence the wavelength is shortened by this distance. The wavelength of the sound received by the observer is $\lambda - v_s T$.

Hence the frequency observed $f_o = v/(\lambda - v_s T) = v/(v/f_s - v_s/f_s)$

$$f_o = \frac{f_s v}{(v - v_s)}$$

The source would move **away** from a stationary observer at position P on the left-hand side of Figure 7.14. The observed wavelengths would lengthen.

For a source of sound moving *away* from an observer the observed frequency can be shown to be

$$f_o = \frac{f_s v}{(v + v_s)}$$

The observed frequency f_o when a source of sound waves moves at speed v_s relative to a stationary observer is

$$f_o = \frac{f_s v}{(v \pm v_s)}$$



The observed frequency is *greater* than the source frequency when the source moves towards the observer and the observed frequency is *less* than the source frequency when the source moves away from the observer.

The above expressions apply only when the source of waves is sound. However, a change of frequency (Doppler shift) is observed with all waves, including light.

EXTENSION

In astronomy, the wavelength tends to be measured rather than the frequency. If the measured wavelength of the emitted light (see Topic 25) is less than that measured for a stationary source, then the distance between the source (star) and detector is decreasing (blueshift). If the measured wavelength is greater than the value of a stationary source, then the source is moving away from the detector (redshift). The blue and red shifts are referred to in this way as red has the longest wavelength in the visible spectrum and blue the shortest.

WORKED EXAMPLE 7D

A police car travels towards a stationary observer at a speed of 15 ms^{-1} . The siren on the car emits a sound of frequency 250 Hz. Calculate the observed frequency. The speed of sound is 340 ms^{-1} .

Answer

$$\begin{aligned} \text{observed frequency } f_o &= \frac{f_s v}{(v - v_s)} \\ &= 250 \times \frac{340}{(340 - 15)} \\ &= 260 \text{ Hz} \end{aligned}$$

Question

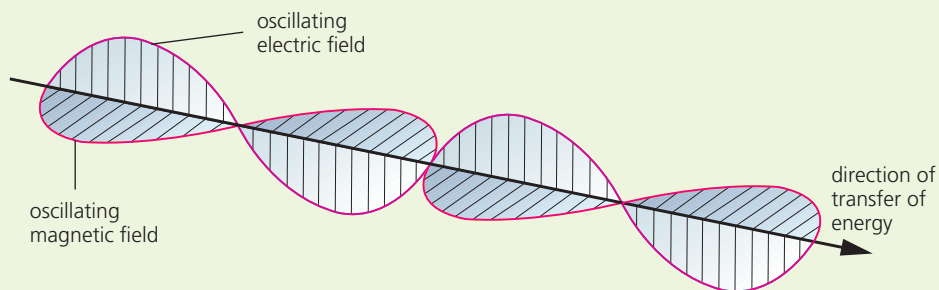
- 7 The sound emitted from the siren of an ambulance has a frequency of 1500 Hz. The speed of sound is 340 ms^{-1} . Calculate the difference in frequency heard by a stationary observer as the ambulance travels towards and then away from the observer at a speed of 30 ms^{-1} .

7.3 Electromagnetic spectrum

Electromagnetic waves are progressive transverse waves (see Topic 7.1). However, unlike other types of transverse wave they do not require a medium to travel through.

EXTENSION

Electromagnetic waves consist of electric and magnetic fields which oscillate at right angles to each other and to the direction in which the wave is travelling. This is illustrated in Figure 7.15.



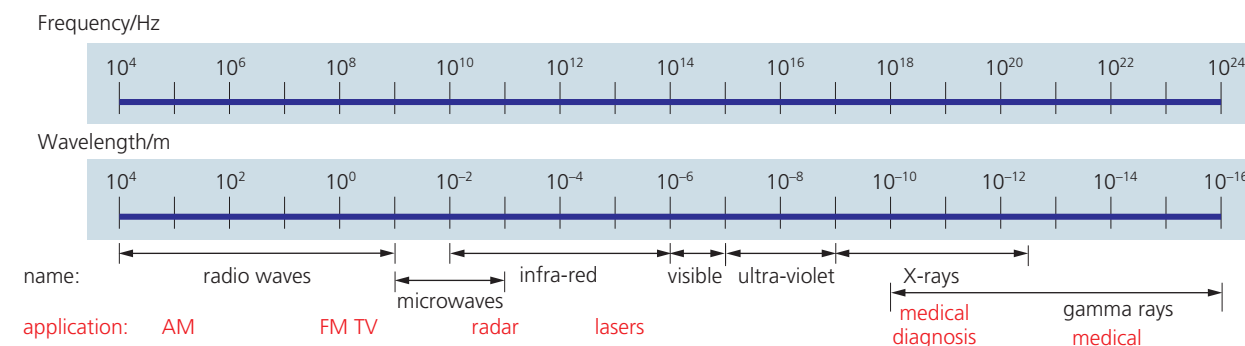
▲ **Figure 7.15** Oscillating electric and magnetic fields in an electromagnetic wave

Electromagnetic (e.m.) waves show all the properties common to wave motions: they can be reflected and refracted. They can also be diffracted and they obey the principle of superposition and produce interference patterns (as we shall see in Topic 8).

In a vacuum (free space) all electromagnetic waves travel at the same speed, $3.00 \times 10^8 \text{ ms}^{-1}$. However their speed is lower in other substances. When a wave slows down, its frequency is unaltered (since this depends on the wave source) but the wavelength will decrease (from $v = f\lambda$).

The complete **electromagnetic spectrum** has a continuous range of frequencies (or wavelengths) but may be divided into a series of regions based on the properties of electromagnetic waves in these regions, as illustrated in Figure 7.16. It should be noted that there is no clear boundary between regions.

Table 7.1 (overleaf) lists the wavelength range of the main regions of the electromagnetic spectrum in free space.



▲ **Figure 7.16** The electromagnetic spectrum

radiation	approximate wavelength range/m
γ -rays	10^{-10} – 10^{-16} and shorter
X-rays	10^{-9} – 10^{-12} and shorter
ultraviolet	10^{-7} – 10^{-9}
visible	4×10^{-7} – 7×10^{-7}
infrared	10^{-2} – 10^{-6}
microwaves	10^{-3} – 10^{-1}
radio waves	10^{-1} – 10^4 and longer

▲ **Table 7.1** Wavelength range of the principal regions of the electromagnetic spectrum in free space

Visible light is just a small region of the electromagnetic spectrum with wavelengths of 400–700 nm in free space, where $1 \text{ nm} = 1 \times 10^{-9} \text{ m}$ (Topic 1). Blue light has a shorter wavelength than red light.

WORKED EXAMPLE 7E

- 1 Calculate the frequency in MHz of a radio wave of wavelength 250 m.
The speed of all electromagnetic waves is $3.00 \times 10^8 \text{ ms}^{-1}$.
- 2 Calculate the wavelength in nm of an X-ray wave of frequency $2.0 \times 10^{18} \text{ Hz}$.

Answers

- 1 For a wave $v = f\lambda$

$$f = 3.00 \times \frac{10^8}{250} = 1.2 \times 10^6 \text{ Hz}$$

$$= \mathbf{1.2 \text{ MHz}}$$
- 2 For a wave $v = f\lambda$

$$\lambda = 3.00 \times 10^8 / 2 \times 10^{18} = 1.5 \times 10^{-10}$$

$$= \mathbf{0.15 \text{ nm}}$$

Questions

- 8 The speed of light is $3.00 \times 10^8 \text{ ms}^{-1}$. Calculate the frequency of red light of wavelength 650 nm. Give your answer in THz.
- 9 A beam of red light has an amplitude that is 2.5 times the amplitude of a second beam of the same colour. Calculate the ratio of the intensities of the waves.
- 10 Calculate the wavelength of microwaves of frequency 8.0 GHz.

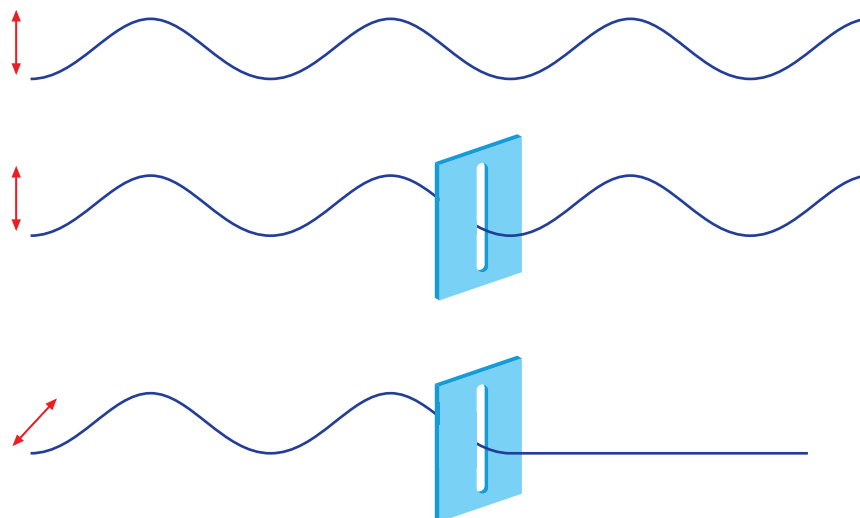


7.4 Polarisation

Consider generating waves in a rope by moving your hand holding the stretched rope up and down, or from side to side. The transverse vibrations of the rope will be in just one plane – the vertical plane if your hand is moving up and down, or the horizontal plane if it moves from side to side. In each case the plane is that containing both the direction of vibration of the rope and the direction in which the wave energy is travelling. These vibrations are said to be **plane polarised** in either a vertical plane or a horizontal plane.

However, there are an infinite number of directions for the vibrations of the rope to be at right angles to the direction the wave energy is travelling (the condition for the wave produced to be a transverse wave). If the direction of the vibration of the rope is continually changed but kept at right angles to the direction of energy travel then

the waves produced are said to be **unpolarised**. This is similar to that shown for unpolarised electromagnetic waves in Figure 7.18.

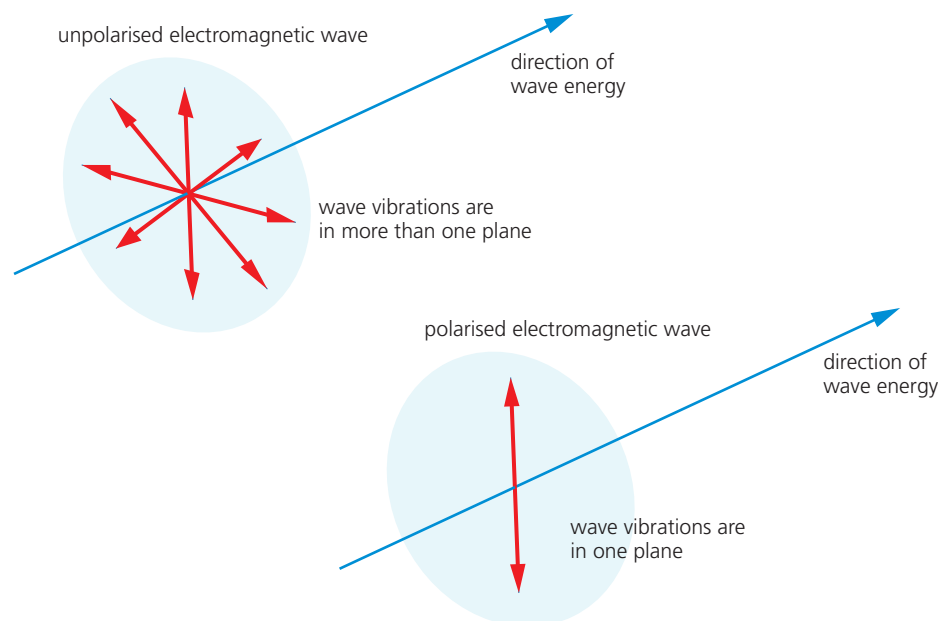


▲ **Figure 7.17** Transverse waves on a rope

A polarised wave is a transverse wave in which vibrations occur in only one of the directions at right angles to the direction in which the wave energy is travelling.

The condition for a wave to be plane polarised is for the vibrations to be in just one plane, which contains the direction in which the wave energy is travelling.

If the rope passes through a vertical slot, then only if you move your hand up and down will the wave pass through the slot. This is illustrated in Figure 7.17. If the rope passes through a horizontal slot, only waves generated by a side-to-side motion of the hand will be transmitted. If the rope passes through a slot of one direction followed by another at right angles, the wave is totally blocked whether it was initially vibrating in the vertical or the horizontal plane. This is similar to that shown for electromagnetic waves in Figure 7.20a on page 125.



▲ **Figure 7.18** Unpolarised and polarised electromagnetic waves

Clearly, polarisation can apply only to transverse waves. In longitudinal waves the vibrations are parallel to the direction of wave travel, and whatever the direction of the slit, it would make no difference to the transmission of the waves.

The Sun and domestic light bulbs emit unpolarised light; that is, over a short period of time, the vibrations take place in many directions in a plane at right angles to the direction of the wave energy, instead of in a single direction of vibration required for plane-polarised electromagnetic waves (see Figure 7.18).

Many radio waves, microwaves and television waves are plane polarised.

Polarising light waves

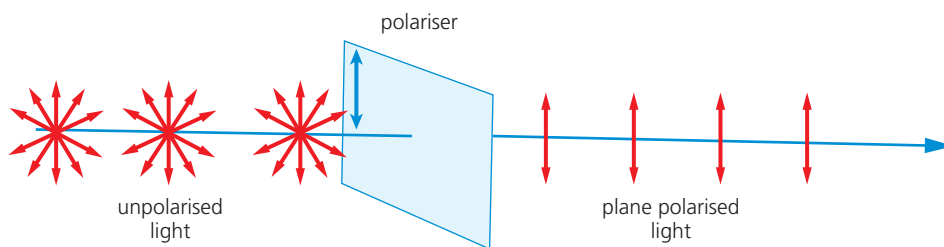
Some transparent materials, such as a Polaroid sheet (used in some types of sunglasses), allow vibrations of light to pass through in one direction only. A Polaroid sheet contains long chains of organic molecules aligned parallel to each other.

EXTENSION

The fact that light can be polarised was understood only in the early 1800s. This was a most important discovery. It showed that light is a transverse wave motion, and opened the way, 50 years later, to Maxwell's theory of light as electromagnetic radiation. James Clerk Maxwell described light in terms of oscillating electric and magnetic fields, at right angles to each other and at right angles to the direction of travel of the wave energy (see Figure 7.15). When we talk about the direction of polarisation of a light wave, we refer to the direction of the electric field component of the electromagnetic wave.

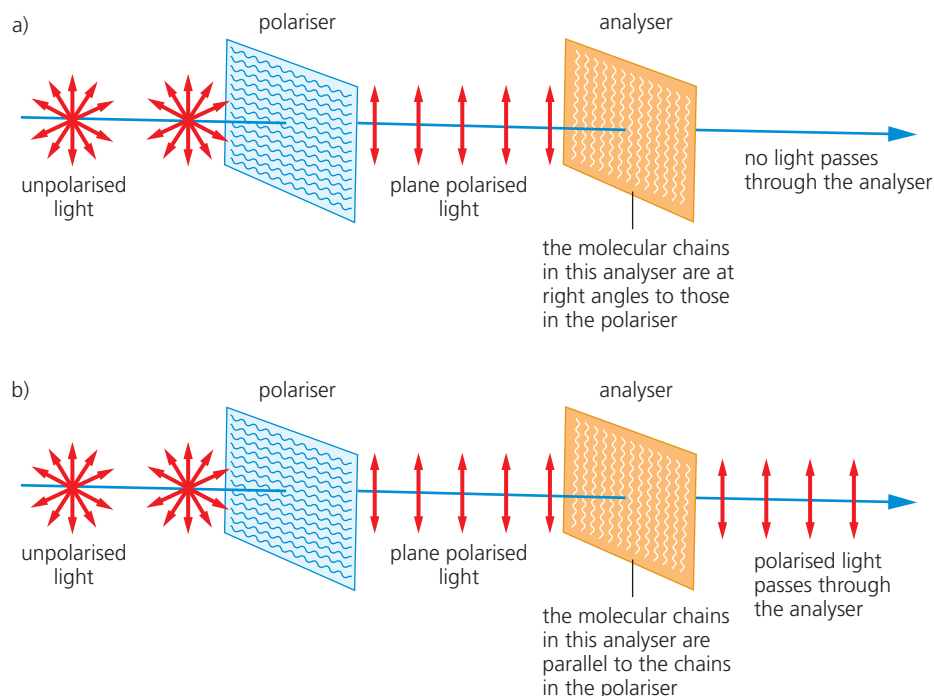
When unpolarised light arrives at a Polaroid sheet, the component of the electric field of the incident light which is parallel to the molecules is strongly absorbed, whereas light with its electric field perpendicular to the molecules is transmitted through the sheet.

The Polaroid sheet acts as a **polariser**, producing plane-polarised light from light that was originally unpolarised. Figure 7.19 illustrates unpolarised light entering the polariser, and polarised light leaving it. The blue arrow in the polariser represents the direction of vibration of the light waves allowed by the polariser.



▲ **Figure 7.19**

If you try to view plane-polarised light through a second sheet of Polaroid which is placed so that its polarising direction is at right angles to the polarising direction of the first sheet, it will be found that no light is transmitted. In this arrangement, the Polaroids are said to be 'crossed'. The second Polaroid sheet is acting as an **analyser**. If the two Polaroids have their polarising directions parallel, then plane-polarised light from the first Polaroid can pass through the second. These two situations are illustrated in Figure 7.20 on the next page. Although the action of the Polaroid sheet is not that of a simple slit, the arrangement of the crossed Polaroids has the same effect as the crossed slits in the rope-and-slits experiment (Figure 7.17).

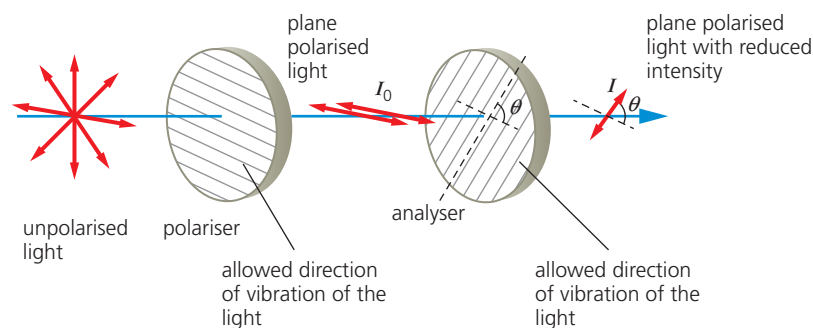


▲ **Figure 7.20** Polariser and analyser in a) crossed and b) parallel situations

Malus's law

When two Polaroid sheets are crossed, none of the polarised light transmitted through the first sheet can be transmitted through the second Polaroid sheet. However, if the two Polaroid sheets are arranged so that the polarising direction of the second sheet is at an angle of less than 90° to the polarising direction of the first sheet, some light can be transmitted. The brightness or intensity of light emerging from the second sheet depends on the angle between the polarising directions of the two sheets.

Consider the maximum light intensity transmitted through the polariser to have amplitude A_0 and intensity I_0 . The plane of polarisation of the light from the polariser is at an angle θ to the polarising direction of the analyser. See Figure 7.21.



▲ **Figure 7.21** Action of an analyser

The analyser polarises the light in a direction parallel to its polarising direction – the plane of polarisation is rotated through an angle θ . The amplitude of the light emerging is $A_0 \cos \theta$. Since intensity is proportional to the amplitude squared ($I \propto A^2$) the transmitted intensity $I \propto A_0^2 \cos^2 \theta$.

and therefore,

$$I = I_0 \cos^2 \theta$$

This is known as **Malus's law**.

Hence, when $\theta = 0$, the maximum intensity I_0 is transmitted, and when $\theta = 90^\circ$, no light is transmitted.

WORKED EXAMPLE 7F

The polariser in Figure 7.19 is slowly rotated through 360° . Describe the intensity of the transmitted plane polarised light as the polariser is rotated.

Answer

The intensity does not vary. (The incident light is unpolarised and, therefore, the vibrations take place in many different directions. On average there are the same number of vibrations in each direction.)

Questions

- 11 The analyser shown in Figure 7.20b is adjusted to give maximum intensity of transmitted light.
Describe the intensity transmitted by the analyser as it is slowly rotated through 360° . Draw a graph to show how the intensity varies with the angle of rotation of the analyser.
- 12 The polariser and analyser have their planes of polarisation parallel (as shown in Figure 7.20b).
Calculate the angle the analyser must be turned through to reduce the transmitted light intensity by a quarter.

SUMMARY

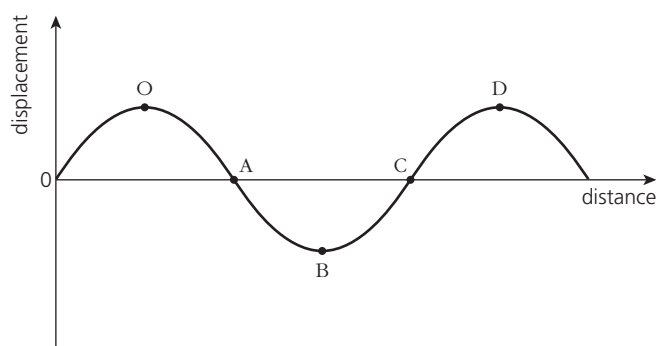
- » A progressive wave travels outwards from the source, carrying energy but without transferring matter.
- » In a transverse wave, the oscillations of the particles are at right angles to the direction in which the wave carries energy.
- » In a longitudinal wave, the oscillations of the particles are in the same direction as the direction in which the wave carries energy.
- » Transverse and longitudinal waves can be represented by a graph of displacement against distance along the wave at a particular moment in time, or as a graph of displacement against time to show how the displacement changes at a particular position.
- » The relative positions of two points on the same wave, or of two waves of the same frequency can be stated as a phase difference in degrees.
- » An oscilloscope can display sound waves as a voltage–time graph from which the frequency and amplitude of the wave can be determined.
- » Properties of wave motion (reflection, refraction, diffraction and interference) can be observed for water waves in a ripple tank.
- » The intensity of a wave is the power per unit area. Intensity is proportional to the square of the amplitude.
- » The speed v , frequency f and wavelength λ of a wave are related by $v = f\lambda$.
- » When a source of sound waves moves relative to a stationary observer there is a change in the observed frequency compared with the frequency emitted by the source. This is the Doppler effect.
- » The observed frequency f_0 is related to the source frequency f and the speed of the source, v , by $f_0 = f_s v / (v \pm v_s)$.
- » All electromagnetic waves have the same speed in free space $c = 3.00 \times 10^8 \text{ m s}^{-1}$.
- » All electromagnetic waves are transverse waves.
- » The wavelengths of electromagnetic waves in free space visible to the human eye are in the range 400 nm to 700 nm.
- » In a plane-polarised wave, the vibrations of the wave are in one direction only, which is at right-angles to the direction of travel of the wave.
- » Transverse waves can be polarised; longitudinal waves cannot.
- » Plane-polarised light can be produced from unpolarised light by using a polariser, such as a sheet of Polaroid.
- » Rotating an analysing Polaroid in a beam of plane-polarised light prevents transmission of the polarised light when the polarising directions of polariser and analyser are at right angles.
- » Malus's law ($I = I_0 \cos^2 \theta$) is used to calculate the intensity of plane-polarised electromagnetic waves after transmission through a polarising filter or series of filters.

END OF TOPIC QUESTIONS

7

End of topic questions

- 1 Fig. 7.22 represents the displacement against distance for a transverse wave. Which point, A, B, C or D vibrates out of phase with point O by 180° ?

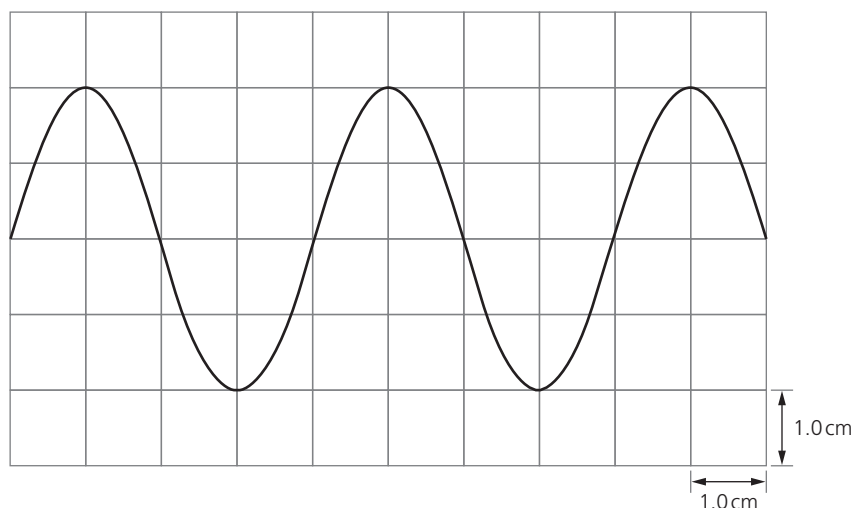


▲ Figure 7.22

- 2 Fig. 7.23 represents the screen of a cathode-ray oscilloscope (c.r.o.). The c.r.o. shows the waveform of a sound wave.

The time-base setting is 0.40 ms cm^{-1} . What is the frequency of the sound wave?

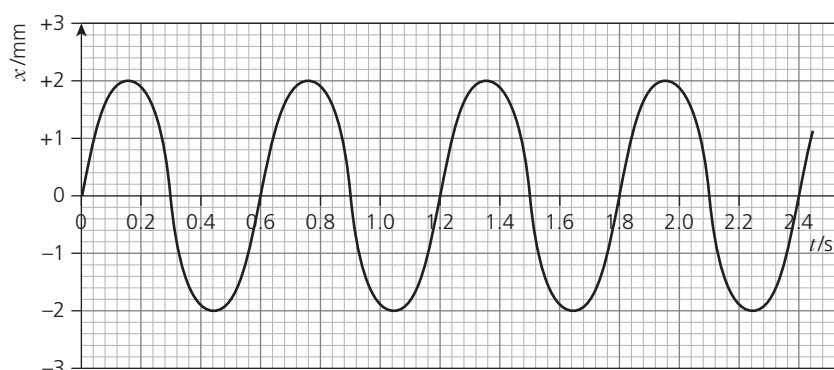
- A 0.63 Hz B 1.3 Hz C 630 Hz D 1300 Hz



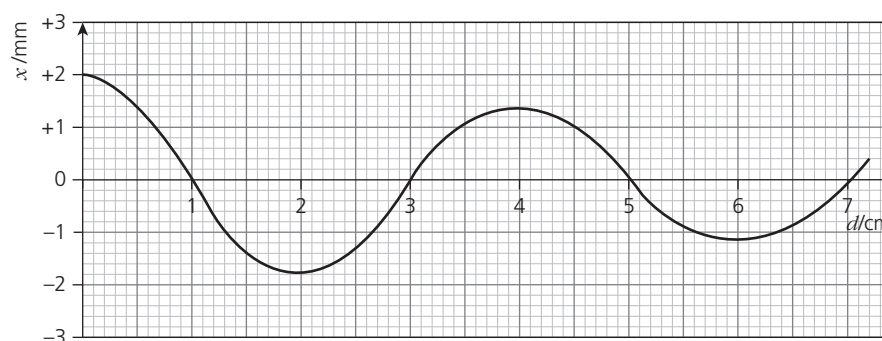
▲ Figure 7.23

- 3 A monochromatic source of electromagnetic waves is viewed through two pieces of polarising filter whose preferred direction of vibration are parallel to each other. What angle should one sheet of polarising filter be turned to reduce the amplitude of the observed wave to half its original value?
- A 30° B 45° C 60° D 90°
- 4 A certain sound wave in air has a speed 340 ms^{-1} and wavelength 1.7 m . For this wave, calculate:
- the frequency,
 - the period.
- 5 The speed of electromagnetic waves in vacuum (or air) is $3.00 \times 10^8 \text{ ms}^{-1}$.
- The visible spectrum extends from a wavelength of 400 nm (blue light) to 700 nm (red light). Calculate the range of frequencies of visible light.
 - A typical frequency for v.h.f. television transmission is 250 MHz . Calculate the corresponding wavelength.
- 6 Two waves travel with the same speed and have the same amplitude, but the first has twice the wavelength of the second. Calculate the ratio of the intensities transmitted by the waves.

- 7 A student stands at a distance of 5.0 m from a point source of sound, which is radiating uniformly in all directions. The intensity of the sound wave at her ear is $6.3 \times 10^{-6} \text{ W m}^{-2}$.
- The receiving area of the student's ear canal is 1.5 cm^2 . Calculate how much energy passes into her ear in 1 minute.
 - The student moves to a point 1.8 m from the source. Calculate the new intensity of the sound.
- 8 Assume that waves spread out uniformly in all directions from the epicentre of an earthquake. The intensity of a particular earthquake wave is measured as $5.0 \times 10^6 \text{ W m}^{-2}$ at a distance of 40 km from the epicentre. What is the intensity at a distance of only 2 km from the epicentre?
- 9 A student is studying a water wave in which all the wavefronts are parallel to one another. The variation with time t of the displacement x of a particular particle in the wave is shown in Fig. 7.24. The distance d of the oscillating particles from the source of the waves is measured. At a particular time, the variation of the displacement x with this distance d is shown in Fig. 7.25.



▲ Figure 7.24

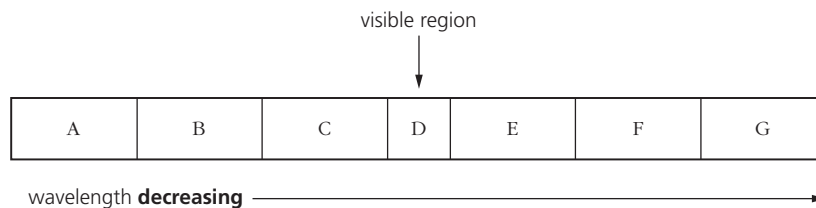


▲ Figure 7.25

- Define, for a wave, what is meant by:
 - displacement, [1]
 - wavelength. [1]
- Use Figs. 7.24 and 7.25 to determine, for the water wave:
 - the period T of vibration, [1]
 - the wavelength λ , [1]
 - the speed v . [2]
- Use Figs. 7.24 and 7.25 to state and explain whether the wave is losing power as it moves away from the source. [2]
 - Determine the ratio $\frac{\text{intensity of wave at source}}{\text{intensity of wave 6.0 cm from source}}$ [3]

Cambridge International AS and A Level Physics (9702) Paper 22 Q5 Oct/Nov 2010

- 10 a** State one property of electromagnetic waves that is **not** common to other transverse waves. [1]
- b** The seven regions of the electromagnetic spectrum are represented by blocks labelled A to G in Fig. 7.26. A typical wavelength for the visible region D is 500 nm. [2]

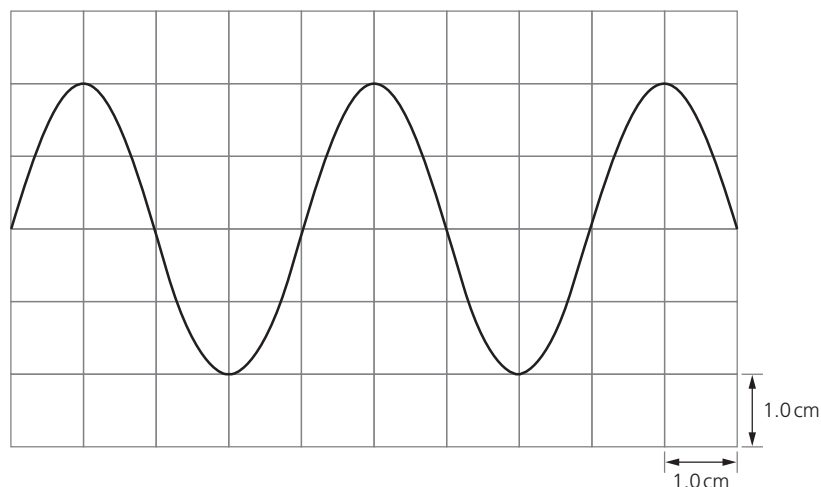


▲ **Figure 7.26**

- i** Name the principle radiations and give a typical wavelength for each of the regions B, E and F. [3]
- ii** Calculate the frequency corresponding to a wavelength of 500 nm. [2]

Cambridge International AS and A Level Physics (9702) Paper 23 Q5 Oct/Nov 2012

- 11 a** A loudspeaker oscillates with frequency f to produce sound waves of wavelength λ . The loudspeaker makes N oscillations in time t .
- i** State expressions, in terms of some or all of the symbols f , λ and N , for:
- 1** the distance moved by a wavefront in time t ,
 - 2** time t .
- ii** Use your answers in **i** to deduce the equation relating the speed v of the sound wave to f and λ . [2]
- b** The waveform of a sound wave is displayed on the screen of a cathode-ray oscilloscope (c.r.o.), as shown in Fig. 7.27. [1]



▲ **Figure 7.27**

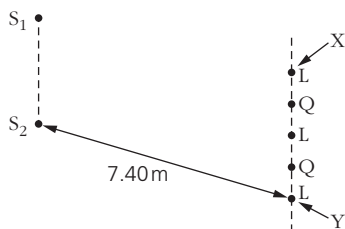
The time-base setting is 0.20 ms cm^{-1} .

Determine the frequency of the sound wave. [2]

- c** Two sources S_1 and S_2 of sound waves are positioned as shown in Fig. 7.28. The sources emit coherent sound waves of wavelength 0.85 m . A sound detector is moved parallel to the line S_1S_2 from a point X to a point Y. Alternate positions of maximum loudness L and minimum loudness Q are detected, as illustrated in Fig. 7.28. Distance S_1X is equal to distance S_2X . Distance S_2Y is 7.40 m .

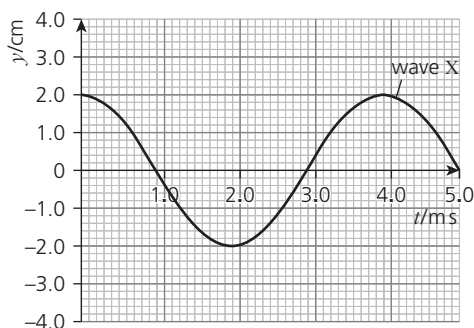
- i** Explain what is meant by *coherent waves*. [1]
- ii** State the phase difference between the two waves arriving at the position of minimum loudness Q that is closest to point X. [1]
- iii** Determine the distance S_1Y . [2]

Cambridge International AS and A Level Physics (9702) Paper 21 Q5 May/June 2019



▲ **Figure 7.28** (not to scale)

- 12 The variation with time t of the displacement y of a wave X, as it passes a point P, is shown in Fig. 7.29.



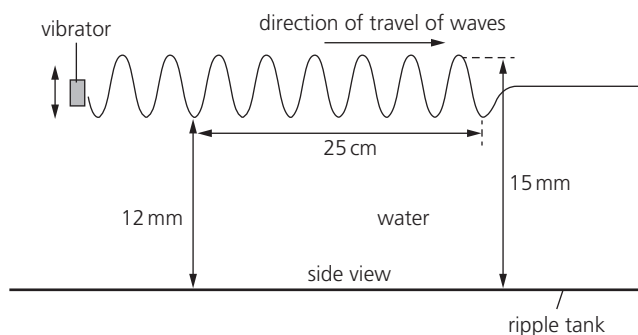
▲ Figure 7.29

The intensity of wave X is I .

- a Use Fig. 7.29 to determine the frequency of wave X. [2]
 b A second wave Z with the same frequency as wave X also passes point P. Wave Z has intensity $2I$. The phase difference between the two waves is 90° . On a copy of Fig. 7.29, sketch the variation with time t of the displacement y of wave Z. Show your working. [3]

Cambridge International AS and A Level Physics (9702) Paper 21 Q5a and b May/June 2016

- 13 a Explain what is meant by the following quantities for a wave on the surface of water: [2]
 i displacement and amplitude, [2]
 ii frequency and time period. [2]
 b Fig. 7.30 represents waves on the surface of water in a ripple tank at one particular instant of time.



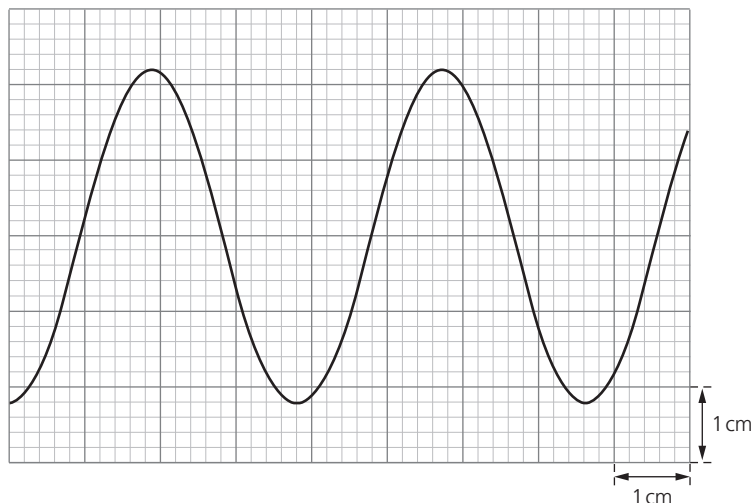
▲ Figure 7.30

A vibrator moves the surface of the water to produce the waves of frequency f . The speed of the waves is 7.5 cm s^{-1} . Where the waves travel on the water surface, the maximum depth of the water is 15 mm and the minimum depth is 12 mm.

- i Calculate, for the waves: [1]
 1 the amplitude, [2]
 2 the wavelength. [2]
 ii Calculate the time period of the oscillations of the vibrator. [2]
 c State and explain whether the waves on the surface of the water shown in Fig. 7.30 are: [1]
 i progressive or stationary, [1]
 ii transverse or longitudinal. [1]

Cambridge International AS and A Level Physics (9702) Paper 21 Q5 May/June 2014

- 14 A microphone detects a musical note of frequency f . The microphone is connected to a cathode-ray oscilloscope (c.r.o.). The signal from the microphone is observed on the c.r.o. as illustrated in Fig. 7.31.



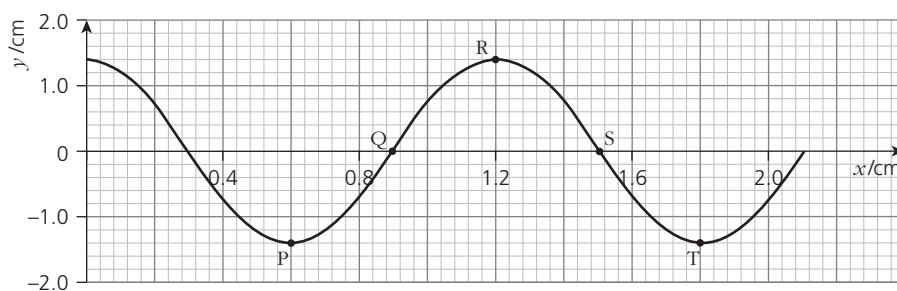
▲ Figure 7.31

The time-base setting of the c.r.o. is 0.50 ms cm^{-1} . The Y-plate setting is 2.5 mV cm^{-1} .

- a Use Fig. 7.31 to determine:
- the amplitude of the signal, [2]
 - the frequency f , [3]
 - the actual uncertainty in f caused by reading the scale on the c.r.o. [2]
- b State f with its actual uncertainty. [1]

Cambridge International AS and A Level Physics (9702) Paper 23 Q2 Oct/Nov 2014

- 15 a i Explain what is meant by a *progressive transverse wave*. [2]
 ii Define *frequency*. [1]
- b The variation with distance x of displacement y for a transverse wave is shown in Fig. 7.32.



▲ Figure 7.32

On Fig. 7.32 five points are labelled. Use Fig. 7.32 to state any two points having a phase difference of:

- zero, [1]
 - 270° . [1]
- c The frequency of the wave in b is 15 Hz. Calculate the speed of the wave in b. [3]
- d Two waves of the same frequency have amplitudes 1.4 cm and 2.1 cm. Calculate the ratio $\frac{\text{intensity of wave of amplitude 1.4 cm}}{\text{intensity of wave of amplitude 2.1 cm}}$ [2]

Cambridge International AS and A Level Physics (9702) Paper 23 Q7 Oct/Nov 2014

- 16 A source of light of one frequency is viewed through two sheets of Polaroid. The preferred directions of vibration of the light for the two sheets of Polaroid are parallel. The intensity of the observed light after passing through the two sheets of Polaroid is I .
- a One sheet of Polaroid is rotated until the amplitude of the vibration of the observed light is reduced to a third of its original value. Calculate the angle rotated by the sheet of Polaroid.
 - b State the effect of this rotation on the intensity of the light transmitted.
 - c Calculate the angle of rotation of one sheet to reduce the intensity of the transmitted light to one third of its original value.
- 17 Light reflected from the surface of smooth water may be described as a polarised transverse wave.
- a By reference to the direction of propagation of energy, explain what is meant by:
 - i a transverse wave, [1]
 - ii polarisation. [1]

Cambridge International AS and A Level Physics (9702) Paper 02 Q5 part a May/June 2007